

Use Case Scenarios - LegalMate AI Project

1. Use Case: User Login s Authentication

Use case name:	User Login s Authentication	
Scenario:	User logs into LegalMate AI platform	
Triggering event:	User wants to access personalized legal services	
Brief description:	User authenticates using credentials, system validates and grants access	
Actors:	User, Admin, Lawyer	
Related use cases:	All system functionalities	
Stakeholders:	System Administrators, Security Team	
Preconditions:	User must have active account, Authentication service available	
Postconditions:	User authenticated, Session created, Access granted based on role	
Flow of activities:	Actor	System
1. User enters email/username and password	1. User provides login credentials	1.1 System validates credentials against User entity 1.2 System checks account status (active/locked) 1.3 System checks role-based permissions
2. User completes authentication	2. User confirms login attempt	2.1 System creates secure session token 2.2 System logs authentication in AdminLog 2.3 System redirects to role-based dashboard
3. User accesses personalized dashboard	3. User views dashboard	3.1 System loads user profile and legal services 3.2 System fetches unread notifications

		3.3 System initializes user context and permissions
Exception conditions:	1.1 Invalid credentials 1.2 Account locked or inactive 2.1 Session creation failed	

2. Use Case: User/Lawyer Registration

Use case name:	User/Lawyer Registration	
Scenario:	New user or lawyer creates an account on LegalMate AI platform	
Triggering event:	New user/lawyer wants to access legal services through the platform	
Brief description:	User/lawyer creates account by selecting role and entering specific information	
Actors:	User (Client), Lawyer	
Related use cases:	Document Analysis, Lawyer Search, Appointment Booking, Lawyer Management	
Stakeholders:	System Administrators, Legal Compliance Team, Bar Association	
Preconditions:	Registration system available, Email service operational, Role verification systems ready	
Postconditions:	Account created and verified, Profile saved, Role-specific dashboard displayed	
Flow of activities:	Actor	System

1. User/Lawyer clicks "Register" and selects role (User/Lawyer)	1. User selects role (User/Lawyer)	1.1 System displays dynamic role-based form 1.2 System validates selected role
2. User/Lawyer enters basic information (name, email, password)	2. User submits name, email, password	2.1 System validates name/email/password 2.2 System checks for duplicate email 2.3 System sends verification email
3. User/Lawyer verifies email through link sent	3. User clicks verification link	3.1 System marks account as verified 3.2 System creates initial profile 3.3 System redirects to profile completion
4. For Users: Completes personal profile	4. User enters personal data	4.1 System saves personal profile 4.2 System displays user dashboard
4. For Lawyers: Completes professional profile (license, specialization, experience)	4. Lawyer submits license C specialization	4.1 System validates license and specialization 4.2 System triggers verification workflow 4.3 System displays "Pending Approval" status
Exception conditions:	1.1 Invalid role selection 2.1 Email already exists 2.2 Weak password 3.1 Verification link expired 4.1 Invalid lawyer license number 4.2 Lawyer verification failed	

3. Use Case: Document Analysis

Use case name:	Document Analysis
Scenario:	User uploads legal document for AI analysis

Triggering event:	User needs to understand legal document content and risks	
Brief description:	User uploads document, AI analyzes content and provides insights	
Actors:	User (Client)	
Related use cases:	Lawyer Recommendation, Risk Assessment	
Stakeholders:	Legal Experts, AI Development Team	
Preconditions:	User must be logged in, Document in supported format, AI services available	
Postconditions:	Document analyzed, Report generated, Recommendations provided	
Flow of activities:	Actor	System
1. User selects "Analyze Document" and uploads file	1. User selects document and uploads	1.1 System validates file type/size 1.2 System stores document securely 1.3 System sends document to AI queue
2. User waits for analysis completion	2. User waits for results	2.1 AI preprocesses document 2.2 AI extracts clauses and risks 2.3 System generates structured analysis report
3. User reviews analysis results	3. User opens analysis report	3.1 System displays report with highlights 3.2 System suggests relevant lawyers 3.3 System saves report to history
Exception conditions:	1.1 Unsupported file format 1.2 File too large 2.1 AI service unavailable 2.2 Poor document quality	

4. Use Case: Lawyer Search s Recommendation

Use case name:	Lawyer Search s Recommendation
Scenario:	User searches for suitable lawyers based on legal needs

Triggering event:	User needs legal representation or consultation	
Brief description:	System recommends lawyers based on specialization, location, and user case	
Actors:	User (Client)	
Related use cases:	Document Analysis, Appointment Booking	
Stakeholders:	Lawyers, Legal Firms, Bar Association	
Preconditions:	User logged in, Lawyer database populated, Location services available	
Postconditions:	Lawyer list generated, Recommendations saved, User can contact lawyers	
Flow of activities:	Actor	System
1. User selects "Find Lawyer" and enters search criteria	1. User submits search preferences	1.1 System fetches all active lawyers 1.2 System filters by specialization/location 1.3 AI ranking algorithm generates matches
2. User reviews lawyer profiles and ratings	2. User browses profiles	2.1 System displays profile + ratings 2.2 System shows availability calendar 2.3 System highlights recommended matches
3. User selects lawyer for consultation	3. User chooses lawyer	3.1 System shows contact/booking options 3.2 System prepares appointment request 3.3 System updates user preference metrics
Exception conditions:	1.1 No lawyers available in area 1.2 No matching specialization 3.1 Lawyer not accepting new clients	

5. Use Case: Appointment Bookings Management

Use case name:	Appointment Bookings Management	
Scenario:	User books appointment with selected lawyer with flexible rescheduling options	
Triggering event:	User wants to schedule legal consultation	
Brief description:	User books appointment, both parties can reschedule with mutual awareness, lawyer manages availability	
Actors:	User (Client), Lawyer	
Related use cases:	Lawyer Search, Notification Management	
Stakeholders:	Lawyers, Legal Assistants, Notification System	
Preconditions:	User logged in, Lawyer selected and available, Consultation system operational	
Postconditions:	Appointment scheduled/rescheduled/cancelled, Notifications sent, Calendar updated	
Flow of activities:	Actor	System
A. Booking Flow		
1. User selects "Book Appointment" with chosen lawyer	1. User chooses time slot	1.1 System loads lawyer availability 1.2 System displays booking policies
2. User selects preferred time and confirms booking	2. User confirms chosen time	2.1 System reserves slot 2.2 System creates Appointment entity 2.3 System sets status = Confirmed
B. Rescheduling by User		
3. User requests to reschedule appointment	3. User opens reschedule option	3.1 System retrieves available slots
4. User confirms new time	4. User selects new time	4.1 System updates appointment 4.2 System notifies lawyer 4.3 System updates calendars

C. Rescheduling by Lawyer		
5. Lawyer requests to reschedule appointment	5. Lawyer submits reschedule request	5.1 System suggests available slots 5.2 System notifies user
6. User approves/rejects new time	6. User responds	6.1 If approved: system updates appointment 6.2 If rejected: prompts lawyer for another slot
D. Lawyer Manages Availability		
7. Lawyer adds new available time slots	7. Lawyer adds new slots	7.1 System saves new slots 7.2 System updates booking engine
Exception conditions:	1.1 No available time slots 3.1 No alternative slots available for rescheduling 5.1 User rejects rescheduling request 6.1 User doesn't respond to rescheduling request	

6. Use Case: Contract Template Generation

Use case name:	Contract Template Generation
Scenario:	User generates legal contract using AI-powered templates
Triggering event:	User needs standardized legal document
Brief description:	AI assists in creating customized legal contracts from templates
Actors:	User (Client)
Related use cases:	Document Analysis, Legal Review
Stakeholders:	Legal Drafting Experts, Compliance Team

Preconditions:	User logged in, Template database available, AI services operational	
Postconditions:	Custom contract generated, User can download and edit	
Flow of activities:	Actor	System
1. User selects contract type and requirements	1. User chooses contract type	1.1 System displays templates 1.2 System prompts for needed data 1.3 System validates legal requirements
2. User inputs specific contract details	2. User fills form fields	2.1 AI adjusts text dynamically 2.2 System inserts user data 2.3 System highlights editable fields
3. User reviews and finalizes contract	3. User reviews and confirms	3.1 System generates PDF/Word 3.2 Provides download options 3.3 Saves contract to history
Exception conditions:	1.1 Unavailable template type 2.1 Incomplete information provided 3.1 Generation failure	

7. Use Case: IR Smart Legal Search

Use case name:	IR Smart Legal Search
Scenario:	User performs intelligent legal document and information search
Triggering event:	User needs to find legal information, documents, or precedents
Brief description:	AI-powered search through legal documents, laws, and articles
Actors:	User, Lawyer
Related use cases:	Document Analysis, Legal Research

Stakeholders:	Legal Researchers, Content Managers	
Preconditions:	Search index available, User authenticated, Sufficient permissions	
Postconditions:	Search results returned, Query logged, Relevant documents identified	
Flow of activities:	Actor	System
1. User enters search query	1. User types search keywords	1.1 System saves IRQueries entry 1.2 AI analyzes query 1.3 System searches SearchIndex
2. System processes and ranks results	2. User waits	2.1 Retrieves IRDocuments 2.2 Applies BM25 + embeddings 2.3 Creates IRQueryDocument relationships
3. User reviews and refines results	3. User browses results	3.1 System shows ranked results 3.2 Provides filters 3.3 Logs search interaction
Exception conditions:	1.1 Search index unavailable 2.1 No results found 3.1 Query too broad or ambiguous	

8. Use Case: Manage Lawyers

Use case name:	Manage Lawyers
Scenario:	Admin manages lawyer profiles and availability
Triggering event:	Lawyer registration or profile updates needed
Brief description:	Administrative management of lawyer accounts, profiles, and status

Actors:	Admin	
Related use cases:	Appointment Management, Consultation Requests	
Stakeholders:	Legal Firms, System Administrators, Clients	
Preconditions:	Admin privileges, Lawyer entity exists, Validation systems available	
Postconditions:	Lawyer profile updated, Status changed, Notifications sent	
Flow of activities:	Actor	System
1. Admin accesses lawyer management	1. Admin accesses dashboard	1.1 System loads Lawyer entities 1.2 Displays management UI 1.3 Loads stats C status
2. Admin performs management actions	2. Admin modifies lawyer data	2.1 System validates updates 2.2 Updates Lawyer entity 2.3 Logs action in AdminLog
3. System propagates changes	3. Admin saves	3.1 Updates SearchIndex 3.2 Sends notifications 3.3 Recalculates availability
Exception conditions:	1.1 Insufficient permissions 2.1 Invalid lawyer data 3.1 Update propagation failed	

9.Use Case: AI Contract Analysis

Use case name:	AI Contract Analysis
Scenario:	User uploads contract for AI-powered analysis and review
Triggering event:	User needs contract reviewed for risks and compliance
Brief description:	AI analyzes contract clauses, identifies risks, suggests improvements
Actors:	User

Related use cases:	IR Smart Legal Search, Consultation Requests	
Stakeholders:	Legal Experts, Compliance Officers, Risk Management	
Preconditions:	Contract uploaded, AI models available, User authenticated	
Postconditions:	Analysis completed, Report generated, Risks identified	
Flow of activities:	Actor	System
1. User uploads contract document	1. User selects contract file	1.1 System saves UserContracts 1.2 Preprocesses text 1.3 Sends to AI pipeline
2. AI analyzes contract content	2. User waits	2.1 Runs through AI models 2.2 Extracts clauses 2.3 Generates ModelResults
3. User receives analysis report	3. User opens analysis	3.1 System compiles report 3.2 Highlights risks 3.3 Stores analysis history
Exception conditions:	1.1 Unsupported document format 2.1 AI service unavailable 3.1 Analysis generation failed	

10. Use Case: AdminLog Management

Use case name:	AdminLog Management
Scenario:	System administrators monitor and manage system activity logs
Triggering event:	Need to review system activities, troubleshoot issues, or generate reports
Brief description:	Admin accesses, filters, and analyzes system logs for monitoring and auditing purposes
Actors:	Admin
Related use cases:	All system functionalities

Stakeholders:	System Administrators, Security Team, Legal Compliance Team	
Preconditions:	Admin must be logged in with appropriate permissions, Log database available	
Postconditions:	Logs accessed and analyzed, Reports generated if needed, Actions taken based on findings	
Flow of activities:	Actor	System
1. Admin accesses AdminLog management interface	1. Admin accesses logs	1.1 System loads AdminLog entities 1.2 Applies default sorting 1.3 Displays dashboard
2. Admin applies filters and searches logs	2. Admin sets filter criteria	2.1 System processes filter query 2.2 Returns filtered results 2.3 Highlights important entries
3. Admin analyzes log data and takes action	3. Admin exports or takes action	3.1 Provides export options 3.2 Generates summary reports 3.3 Allows bulk actions 3.4 Logs admin's access
Exception conditions:	1.1 Insufficient permissions 2.1 No logs match criteria 3.1 Export generation failed 3.2 Database connection issues	